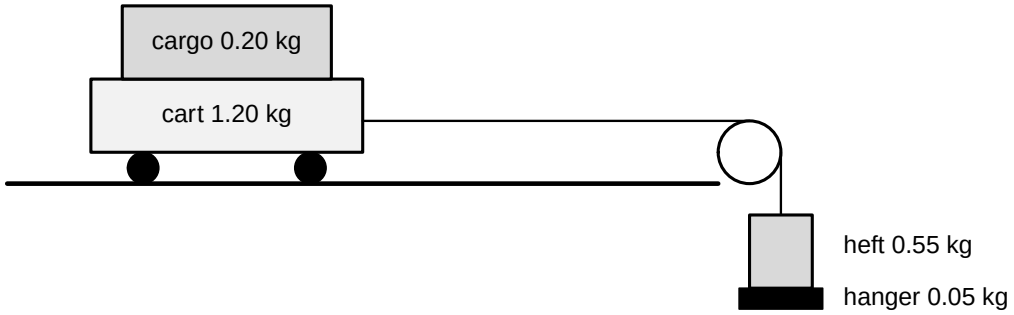
1. **Problem**
   A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



   - The **forward force** is the weight of the hanger plus the heft.
   - Weight is found by multiplying mass by 10 N/kg.
   - The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

   What is the forward force (include units!)?

   What is the total mass (include units!)?

   What is the acceleration of the cart (include units!)?

   Solution

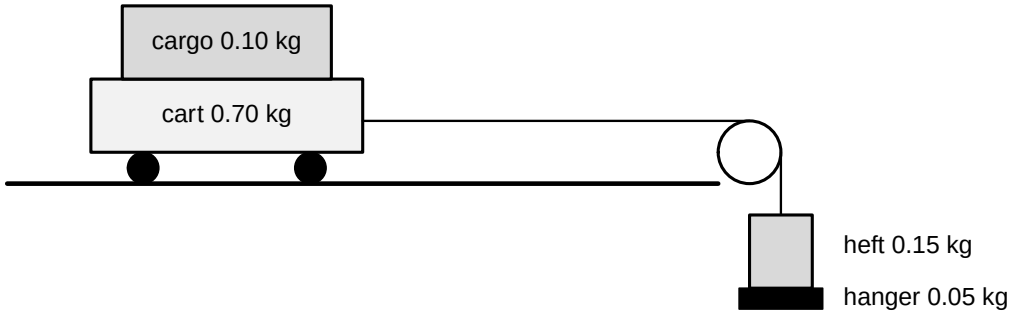
   The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.55 = 0.60$ kg, so the weight is $0.60 \times 10 = 6$ N.

   The total mass includes all four masses: $1.20 + 0.20 + 0.05 + 0.55 = 2.00$ kg.

   The acceleration is the force divided by the total mass: $6 \div 2 = 3$ m/s^2.

   - $(0.05 + 0.55) \times 10 = 6$ N
   - $1.20 + 0.20 + 0.05 + 0.55 = 2.00$ kg
   - $6 \div 2 = 3$ m/s^2

2. **Problem**
A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.15 = 0.20$ kg, so the weight is $0.20 \times 10 = 2$ N.

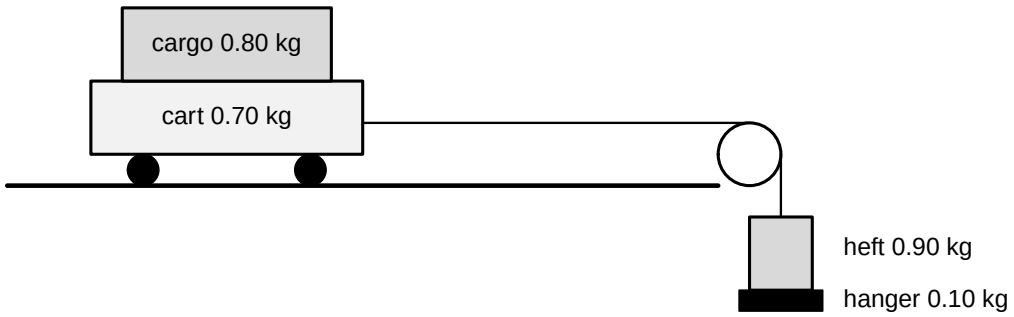
The total mass includes all four masses: $0.70 + 0.10 + 0.05 + 0.15 = 1.00$ kg.

The acceleration is the force divided by the total mass: $2 \div 1 = 2$ m/s^2.

- $(0.05 + 0.15) \times 10 = 2$ N
- $0.70 + 0.10 + 0.05 + 0.15 = 1.00$ kg
- $2 \div 1 = 2$ m/s^2

3. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

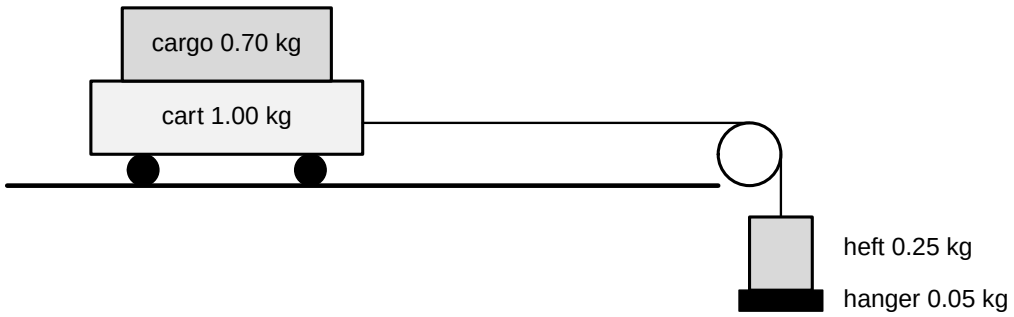
The forward force is the weight of the hanger plus the heft. Their total mass is $0.10 + 0.90 = 1.00$ kg, so the weight is $1.00 \times 10 = 10$ N.

The total mass includes all four masses: $0.70 + 0.80 + 0.10 + 0.90 = 2.50$ kg.

The acceleration is the force divided by the total mass: $10 \div 2.5 = 4$ m/s^2.

- $(0.10 + 0.90) \times 10 = 10$ N
- $0.70 + 0.80 + 0.10 + 0.90 = 2.50$ kg
- $10 \div 2.5 = 4$ m/s^2

4. **Problem**
A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

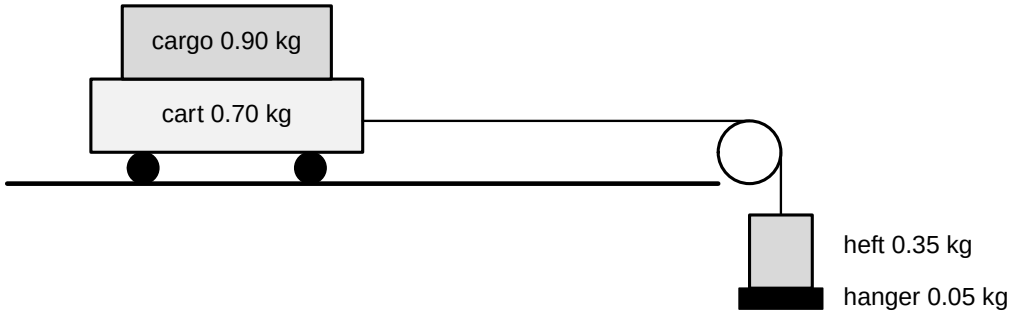
The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.25 = 0.30$ kg, so the weight is $0.30 \times 10 = 3$ N.

The total mass includes all four masses: $1.00 + 0.70 + 0.05 + 0.25 = 2.00$ kg.

The acceleration is the force divided by the total mass: $3 \div 2 = 1.5$ m/s^2.

- $(0.05 + 0.25) \times 10 = 3$ N
- $1.00 + 0.70 + 0.05 + 0.25 = 2.00$ kg
- $3 \div 2 = 1.5$ m/s^2

5. **Problem**
A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

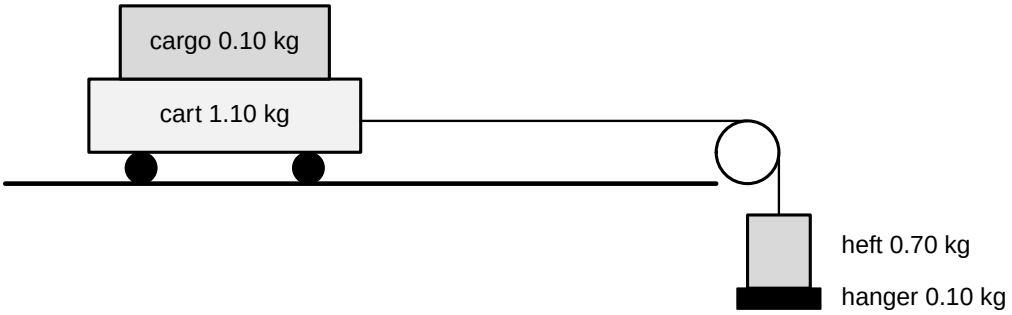
The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.35 = 0.40$ kg, so the weight is $0.40 \times 10 = 4$ N.

The total mass includes all four masses: $0.70 + 0.90 + 0.05 + 0.35 = 2.00$ kg.

The acceleration is the force divided by the total mass: $4 \div 2 = 2$ m/s^2.

- $(0.05 + 0.35) \times 10 = 4$ N
- $0.70 + 0.90 + 0.05 + 0.35 = 2.00$ kg
- $4 \div 2 = 2$ m/s^2

6. **Problem**
A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.10 + 0.70 = 0.80$ kg, so the weight is $0.80 \times 10 = 8$ N.

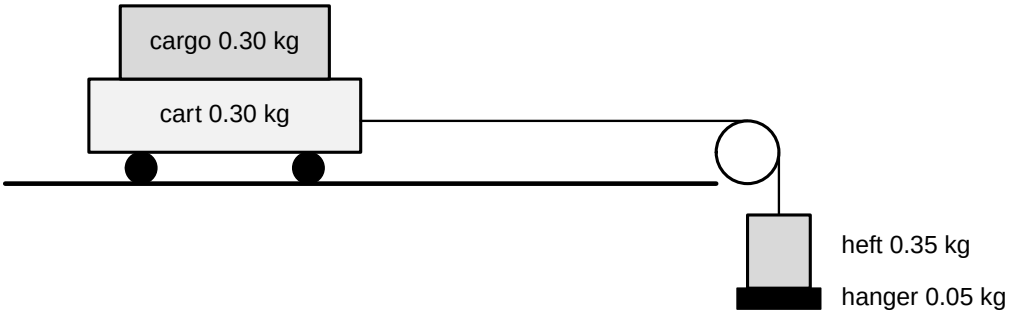
The total mass includes all four masses: $1.10 + 0.10 + 0.10 + 0.70 = 2.00$ kg.

The acceleration is the force divided by the total mass: $8 \div 2 = 4$ m/s^2.

- $(0.10 + 0.70) \times 10 = 8$ N
- $1.10 + 0.10 + 0.10 + 0.70 = 2.00$ kg
- $8 \div 2 = 4$ m/s^2

7. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

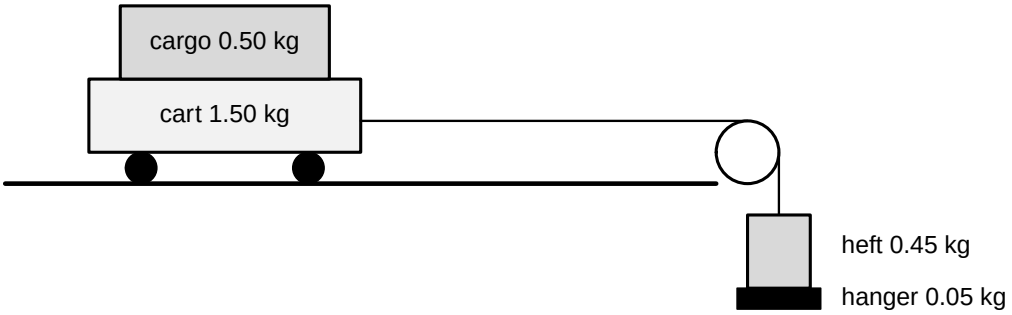
The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.35 = 0.40$ kg, so the weight is $0.40 \times 10 = 4$ N.

The total mass includes all four masses: $0.30 + 0.30 + 0.05 + 0.35 = 1.00$ kg.

The acceleration is the force divided by the total mass: $4 \div 1 = 4$ m/s^2.

- $(0.05 + 0.35) \times 10 = 4$ N
- $0.30 + 0.30 + 0.05 + 0.35 = 1.00$ kg
- $4 \div 1 = 4$ m/s^2

8. **Problem**
A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.45 = 0.50$ kg, so the weight is $0.50 \times 10 = 5$ N.

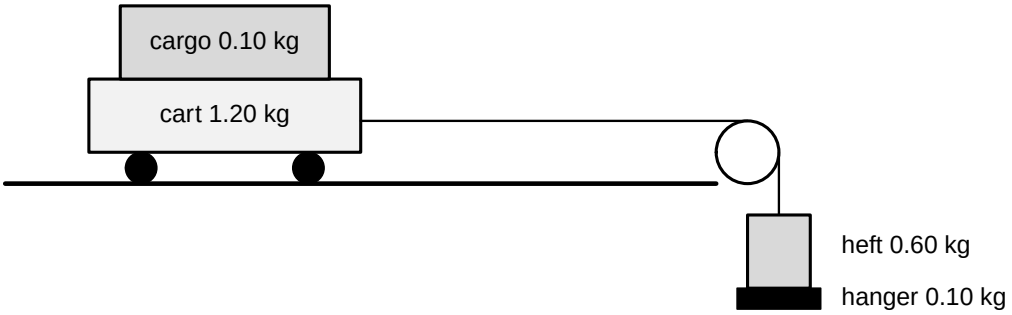
The total mass includes all four masses: $1.50 + 0.50 + 0.05 + 0.45 = 2.50$ kg.

The acceleration is the force divided by the total mass: $5 \div 2.5 = 2$ m/s^2.

- $(0.05 + 0.45) \times 10 = 5$ N
- $1.50 + 0.50 + 0.05 + 0.45 = 2.50$ kg
- $5 \div 2.5 = 2$ m/s^2

9. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.10 + 0.60 = 0.70$ kg, so the weight is $0.70 \times 10 = 7$ N.

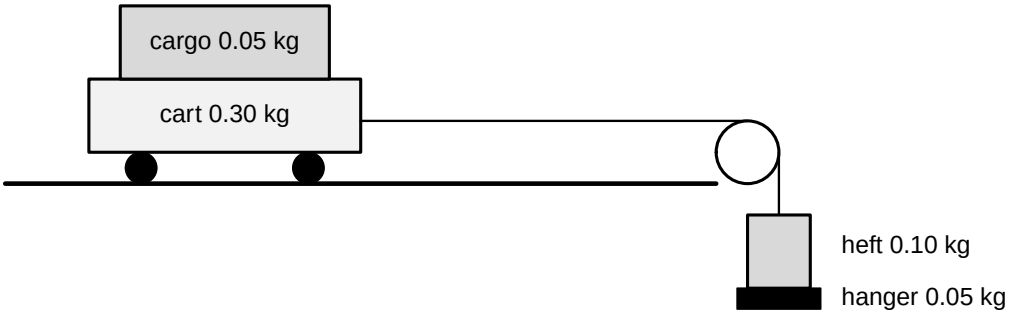
The total mass includes all four masses: $1.20 + 0.10 + 0.10 + 0.60 = 2.00$ kg.

The acceleration is the force divided by the total mass: $7 \div 2 = 3.5$ m/s^2.

- $(0.10 + 0.60) \times 10 = 7$ N
- $1.20 + 0.10 + 0.10 + 0.60 = 2.00$ kg
- $7 \div 2 = 3.5$ m/s^2

10. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.10 = 0.15$ kg, so the weight is $0.15 \times 10 = 1.5$ N.

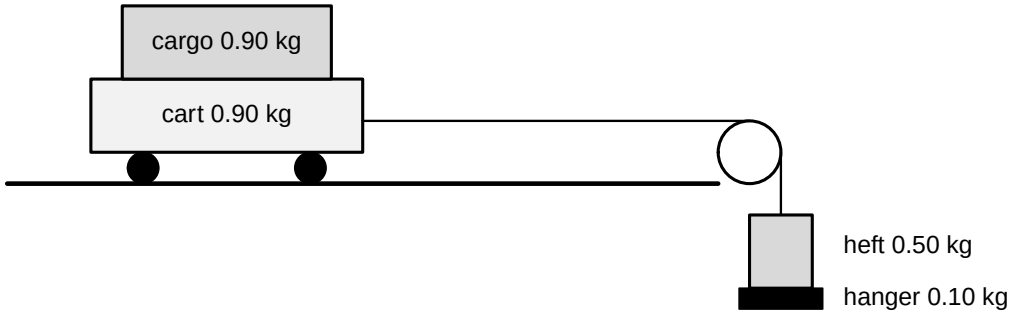
The total mass includes all four masses: $0.30 + 0.05 + 0.05 + 0.10 = 0.50$ kg.

The acceleration is the force divided by the total mass: $1.5 \div 0.5 = 3$ m/s^2.

- $(0.05 + 0.10) \times 10 = 1.5$ N
- $0.30 + 0.05 + 0.05 + 0.10 = 0.50$ kg
- $1.5 \div 0.5 = 3$ m/s^2

11. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.10 + 0.50 = 0.60$ kg, so the weight is $0.60 \times 10 = 6$ N.

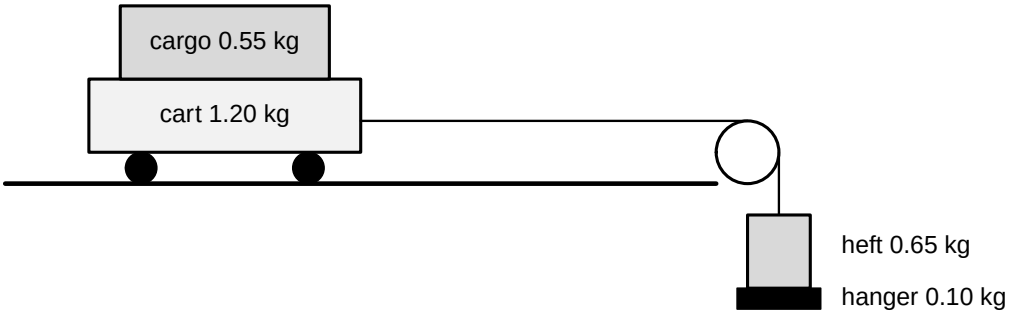
The total mass includes all four masses: $0.90 + 0.90 + 0.10 + 0.50 = 2.40$ kg.

The acceleration is the force divided by the total mass: $6 \div 2.4 = 2.5$ m/s^2.

- $(0.10 + 0.50) \times 10 = 6$ N
- $0.90 + 0.90 + 0.10 + 0.50 = 2.40$ kg
- $6 \div 2.4 = 2.5$ m/s^2

12. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.10 + 0.65 = 0.75$ kg, so the weight is $0.75 \times 10 = 7.5$ N.

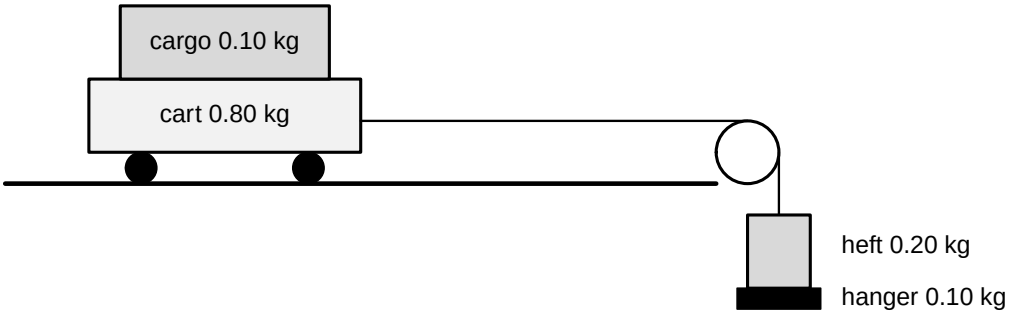
The total mass includes all four masses: $1.20 + 0.55 + 0.10 + 0.65 = 2.50$ kg.

The acceleration is the force divided by the total mass: $7.5 \div 2.5 = 3$ m/s^2.

- $(0.10 + 0.65) \times 10 = 7.5$ N
- $1.20 + 0.55 + 0.10 + 0.65 = 2.50$ kg
- $7.5 \div 2.5 = 3$ m/s^2

13. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.10 + 0.20 = 0.30$ kg, so the weight is $0.30 \times 10 = 3$ N.

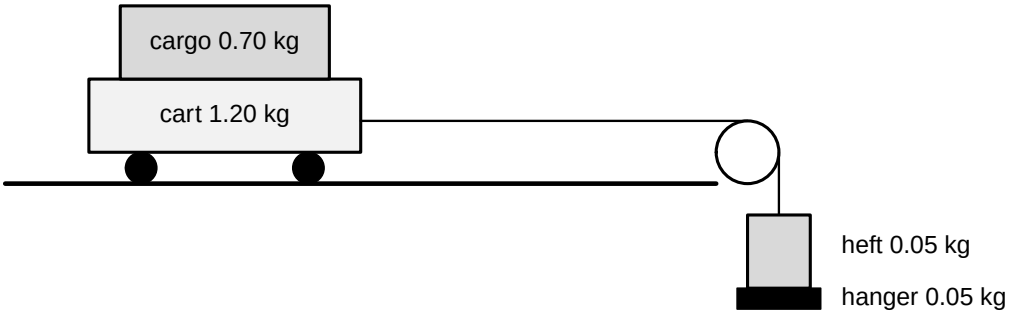
The total mass includes all four masses: $0.80 + 0.10 + 0.10 + 0.20 = 1.20$ kg.

The acceleration is the force divided by the total mass: $3 \div 1.2 = 2.5$ m/s^2.

- $(0.10 + 0.20) \times 10 = 3$ N
- $0.80 + 0.10 + 0.10 + 0.20 = 1.20$ kg
- $3 \div 1.2 = 2.5$ m/s^2

14. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.05 = 0.10$ kg, so the weight is $0.10 \times 10 = 1$ N.

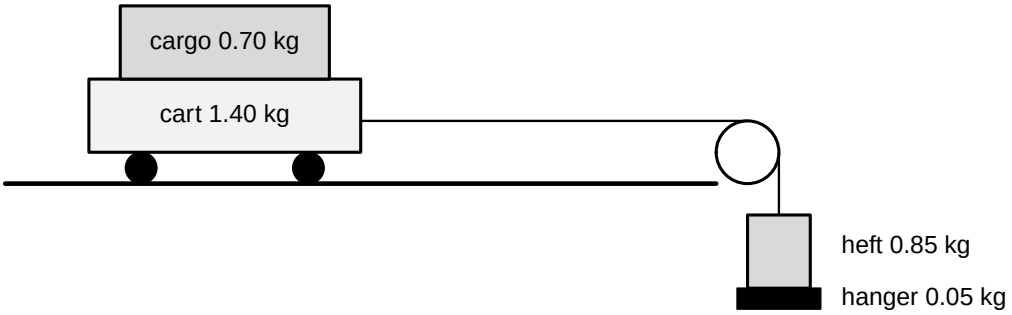
The total mass includes all four masses: $1.20 + 0.70 + 0.05 + 0.05 = 2.00$ kg.

The acceleration is the force divided by the total mass: $1 \div 2 = 0.5$ m/s^2.

- $(0.05 + 0.05) \times 10 = 1$ N
- $1.20 + 0.70 + 0.05 + 0.05 = 2.00$ kg
- $1 \div 2 = 0.5$ m/s^2

15. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.85 = 0.90$ kg, so the weight is $0.90 \times 10 = 9$ N.

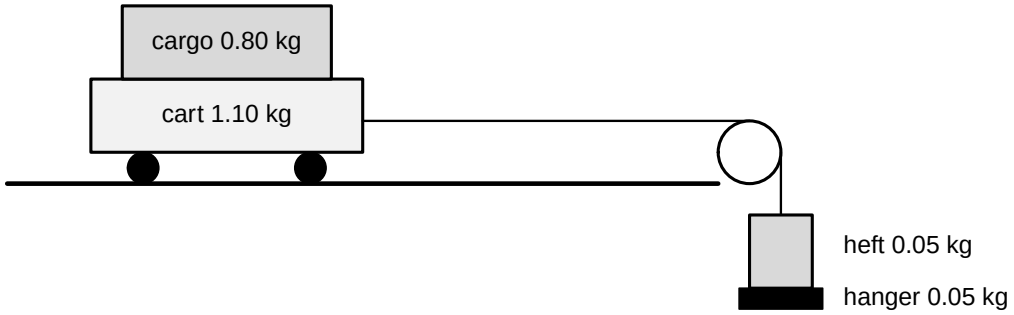
The total mass includes all four masses: $1.40 + 0.70 + 0.05 + 0.85 = 3.00$ kg.

The acceleration is the force divided by the total mass: $9 \div 3 = 3$ m/s^2.

- $(0.05 + 0.85) \times 10 = 9$ N
- $1.40 + 0.70 + 0.05 + 0.85 = 3.00$ kg
- $9 \div 3 = 3$ m/s^2

16. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.05 = 0.10$ kg, so the weight is $0.10 \times 10 = 1$ N.

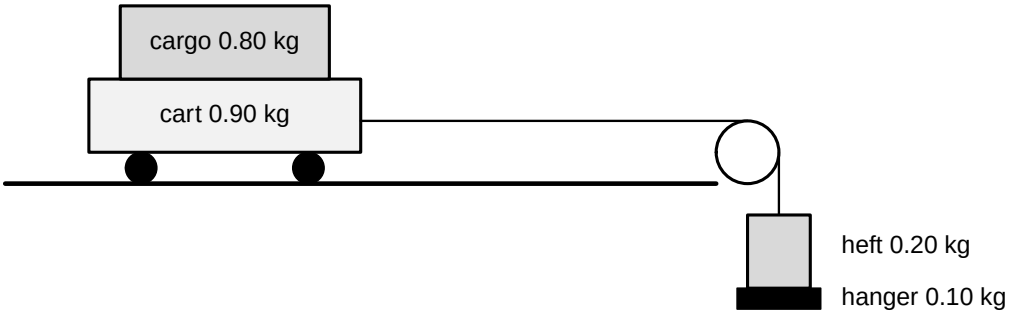
The total mass includes all four masses: $1.10 + 0.80 + 0.05 + 0.05 = 2.00$ kg.

The acceleration is the force divided by the total mass: $1 \div 2 = 0.5$ m/s^2.

- $(0.05 + 0.05) \times 10 = 1$ N
- $1.10 + 0.80 + 0.05 + 0.05 = 2.00$ kg
- $1 \div 2 = 0.5$ m/s^2

17. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.10 + 0.20 = 0.30$ kg, so the weight is $0.30 \times 10 = 3$ N.

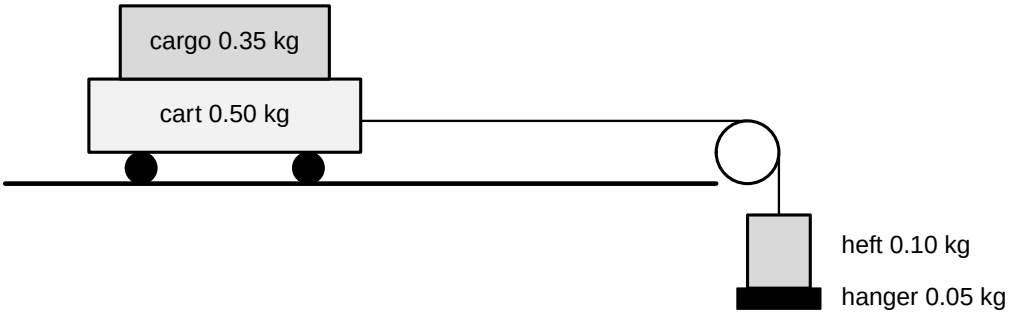
The total mass includes all four masses: $0.90 + 0.80 + 0.10 + 0.20 = 2.00$ kg.

The acceleration is the force divided by the total mass: $3 \div 2 = 1.5$ m/s^2.

- $(0.10 + 0.20) \times 10 = 3$ N
- $0.90 + 0.80 + 0.10 + 0.20 = 2.00$ kg
- $3 \div 2 = 1.5$ m/s^2

18. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.10 = 0.15$ kg, so the weight is $0.15 \times 10 = 1.5$ N.

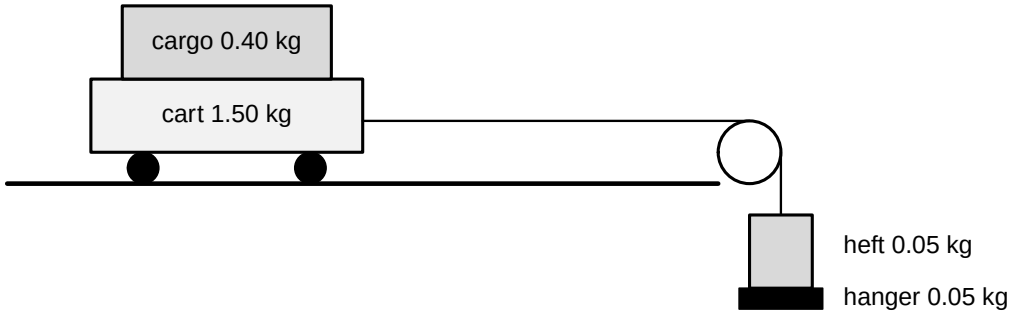
The total mass includes all four masses: $0.50 + 0.35 + 0.05 + 0.10 = 1.00$ kg.

The acceleration is the force divided by the total mass: $1.5 \div 1 = 1.5$ m/s^2.

- $(0.05 + 0.10) \times 10 = 1.5$ N
- $0.50 + 0.35 + 0.05 + 0.10 = 1.00$ kg
- $1.5 \div 1 = 1.5$ m/s^2

19. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.05 = 0.10$ kg, so the weight is $0.10 \times 10 = 1$ N.

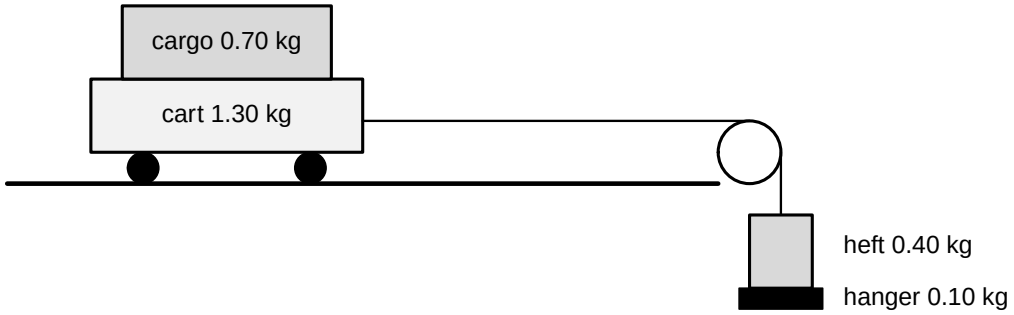
The total mass includes all four masses: $1.50 + 0.40 + 0.05 + 0.05 = 2.00$ kg.

The acceleration is the force divided by the total mass: $1 \div 2 = 0.5$ m/s^2.

- $(0.05 + 0.05) \times 10 = 1$ N
- $1.50 + 0.40 + 0.05 + 0.05 = 2.00$ kg
- $1 \div 2 = 0.5$ m/s^2

20. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.10 + 0.40 = 0.50$ kg, so the weight is $0.50 \times 10 = 5$ N.

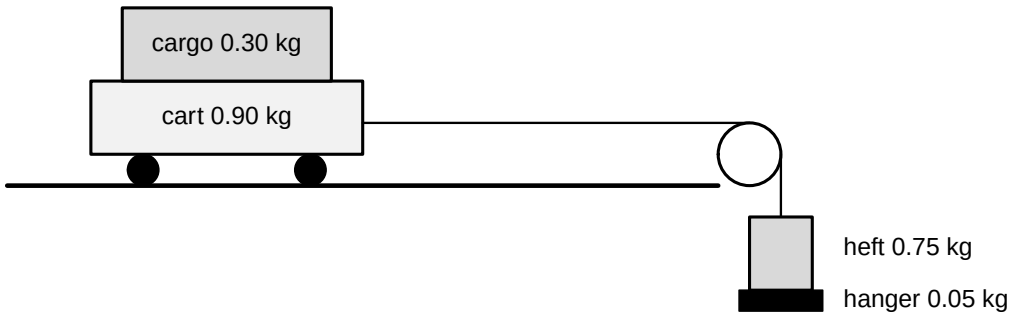
The total mass includes all four masses: $1.30 + 0.70 + 0.10 + 0.40 = 2.50$ kg.

The acceleration is the force divided by the total mass: $5 \div 2.5 = 2$ m/s^2.

- $(0.10 + 0.40) \times 10 = 5$ N
- $1.30 + 0.70 + 0.10 + 0.40 = 2.50$ kg
- $5 \div 2.5 = 2$ m/s^2

21. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.75 = 0.80$ kg, so the weight is $0.80 \times 10 = 8$ N.

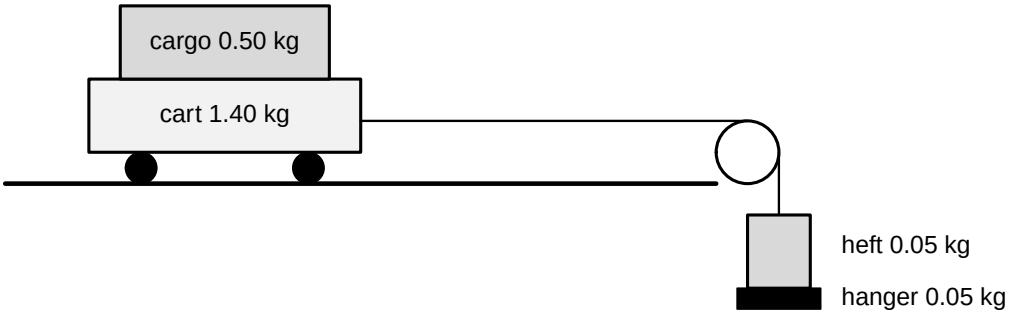
The total mass includes all four masses: $0.90 + 0.30 + 0.05 + 0.75 = 2.00$ kg.

The acceleration is the force divided by the total mass: $8 \div 2 = 4$ m/s^2.

- $(0.05 + 0.75) \times 10 = 8$ N
- $0.90 + 0.30 + 0.05 + 0.75 = 2.00$ kg
- $8 \div 2 = 4$ m/s^2

22. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.05 = 0.10$ kg, so the weight is $0.10 \times 10 = 1$ N.

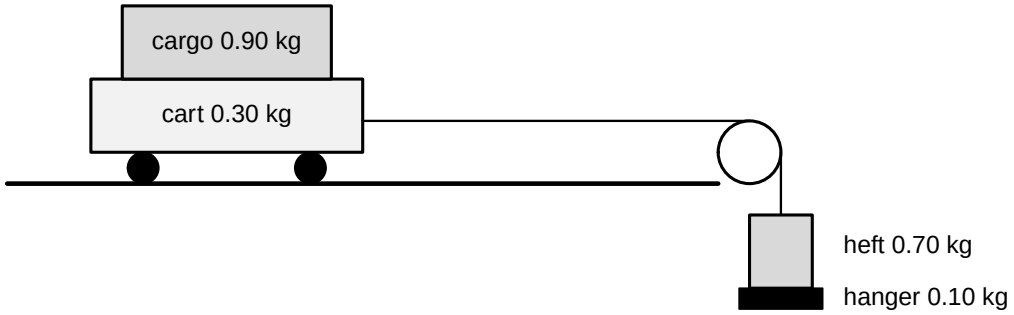
The total mass includes all four masses: $1.40 + 0.50 + 0.05 + 0.05 = 2.00$ kg.

The acceleration is the force divided by the total mass: $1 \div 2 = 0.5$ m/s^2.

- $(0.05 + 0.05) \times 10 = 1$ N
- $1.40 + 0.50 + 0.05 + 0.05 = 2.00$ kg
- $1 \div 2 = 0.5$ m/s^2

23. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.10 + 0.70 = 0.80$ kg, so the weight is $0.80 \times 10 = 8$ N.

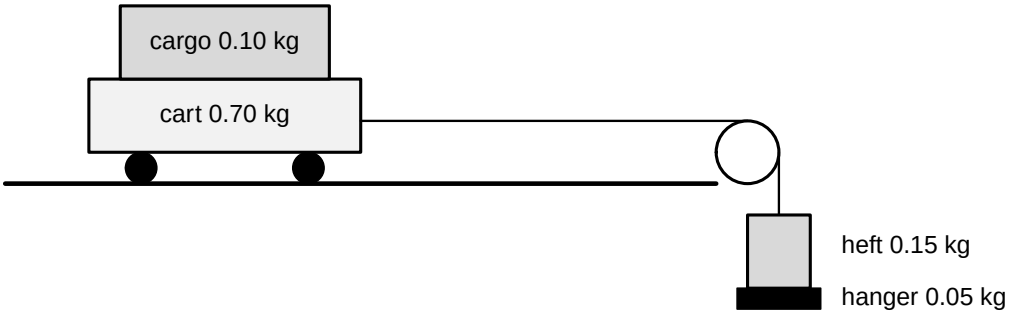
The total mass includes all four masses: $0.30 + 0.90 + 0.10 + 0.70 = 2.00$ kg.

The acceleration is the force divided by the total mass: $8 \div 2 = 4$ m/s^2.

- $(0.10 + 0.70) \times 10 = 8$ N
- $0.30 + 0.90 + 0.10 + 0.70 = 2.00$ kg
- $8 \div 2 = 4$ m/s^2

24. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.15 = 0.20$ kg, so the weight is $0.20 \times 10 = 2$ N.

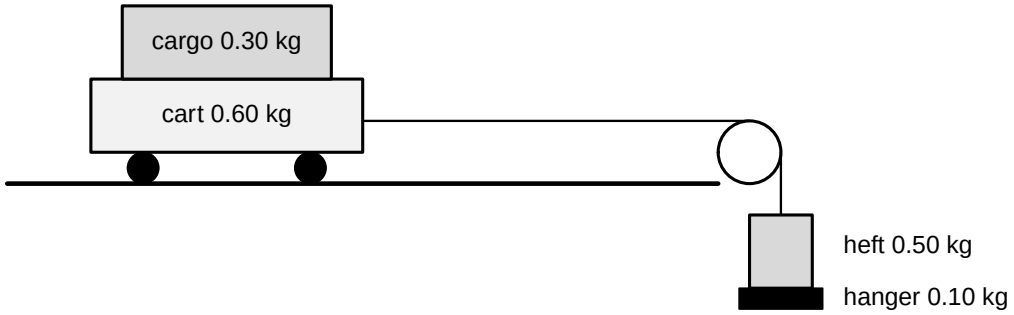
The total mass includes all four masses: $0.70 + 0.10 + 0.05 + 0.15 = 1.00$ kg.

The acceleration is the force divided by the total mass: $2 \div 1 = 2$ m/s^2.

- $(0.05 + 0.15) \times 10 = 2$ N
- $0.70 + 0.10 + 0.05 + 0.15 = 1.00$ kg
- $2 \div 1 = 2$ m/s^2

25. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.10 + 0.50 = 0.60$ kg, so the weight is $0.60 \times 10 = 6$ N.

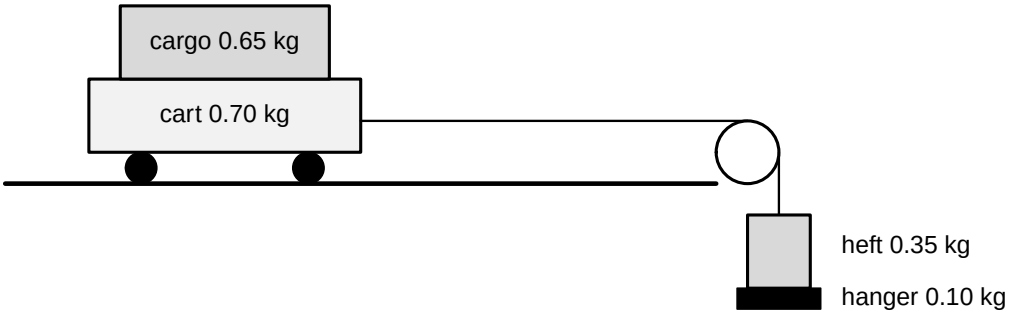
The total mass includes all four masses: $0.60 + 0.30 + 0.10 + 0.50 = 1.50$ kg.

The acceleration is the force divided by the total mass: $6 \div 1.5 = 4$ m/s^2.

- $(0.10 + 0.50) \times 10 = 6$ N
- $0.60 + 0.30 + 0.10 + 0.50 = 1.50$ kg
- $6 \div 1.5 = 4$ m/s^2

26. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.10 + 0.35 = 0.45$ kg, so the weight is $0.45 \times 10 = 4.5$ N.

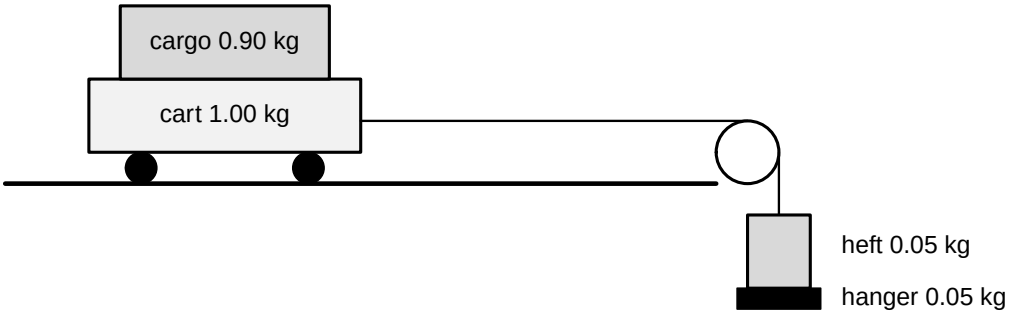
The total mass includes all four masses: $0.70 + 0.65 + 0.10 + 0.35 = 1.80$ kg.

The acceleration is the force divided by the total mass: $4.5 \div 1.8 = 2.5$ m/s^2.

- $(0.10 + 0.35) \times 10 = 4.5$ N
- $0.70 + 0.65 + 0.10 + 0.35 = 1.80$ kg
- $4.5 \div 1.8 = 2.5$ m/s^2

27. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.05 = 0.10$ kg, so the weight is $0.10 \times 10 = 1$ N.

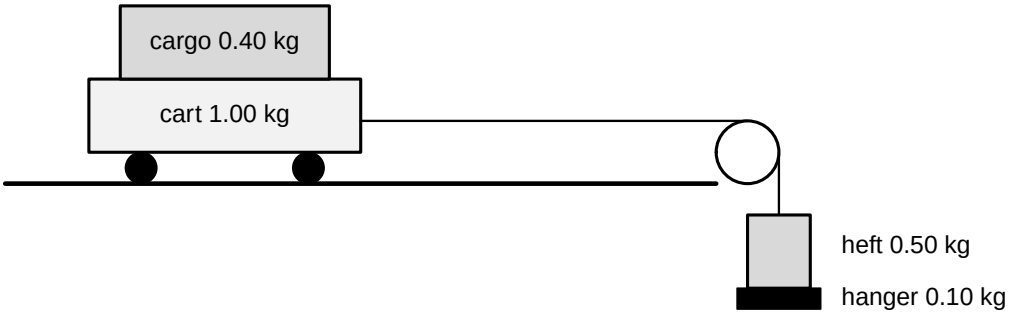
The total mass includes all four masses: $1.00 + 0.90 + 0.05 + 0.05 = 2.00$ kg.

The acceleration is the force divided by the total mass: $1 \div 2 = 0.5$ m/s^2.

- $(0.05 + 0.05) \times 10 = 1$ N
- $1.00 + 0.90 + 0.05 + 0.05 = 2.00$ kg
- $1 \div 2 = 0.5$ m/s^2

28. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.10 + 0.50 = 0.60$ kg, so the weight is $0.60 \times 10 = 6$ N.

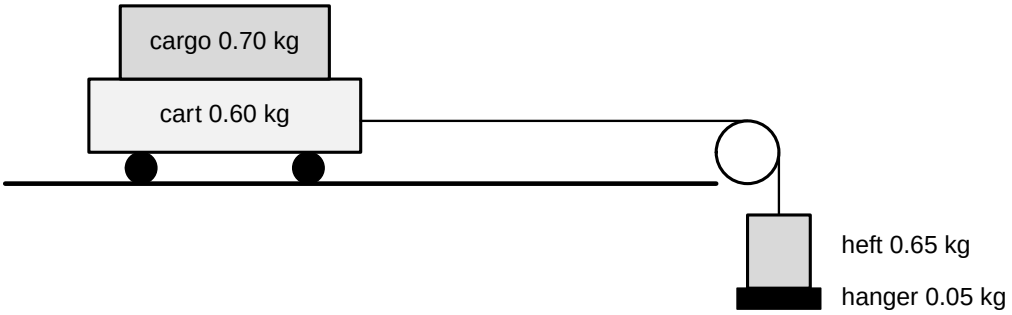
The total mass includes all four masses: $1.00 + 0.40 + 0.10 + 0.50 = 2.00$ kg.

The acceleration is the force divided by the total mass: $6 \div 2 = 3$ m/s^2.

- $(0.10 + 0.50) \times 10 = 6$ N
- $1.00 + 0.40 + 0.10 + 0.50 = 2.00$ kg
- $6 \div 2 = 3$ m/s^2

29. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.65 = 0.70$ kg, so the weight is $0.70 \times 10 = 7$ N.

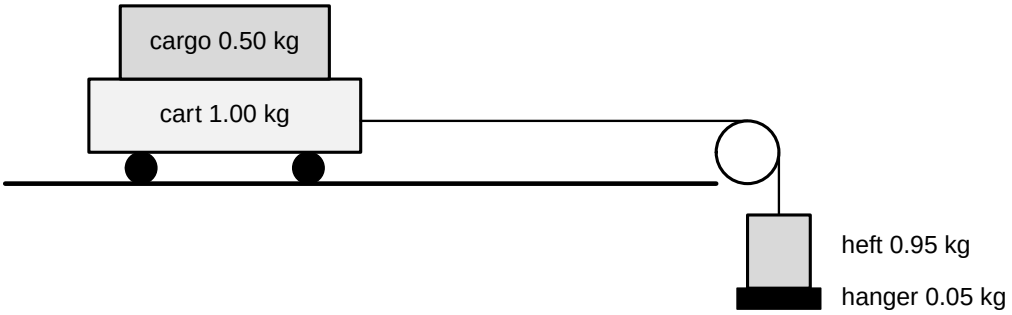
The total mass includes all four masses: $0.60 + 0.70 + 0.05 + 0.65 = 2.00$ kg.

The acceleration is the force divided by the total mass: $7 \div 2 = 3.5$ m/s^2.

- $(0.05 + 0.65) \times 10 = 7$ N
- $0.60 + 0.70 + 0.05 + 0.65 = 2.00$ kg
- $7 \div 2 = 3.5$ m/s^2

30. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.95 = 1.00$ kg, so the weight is $1.00 \times 10 = 10$ N.

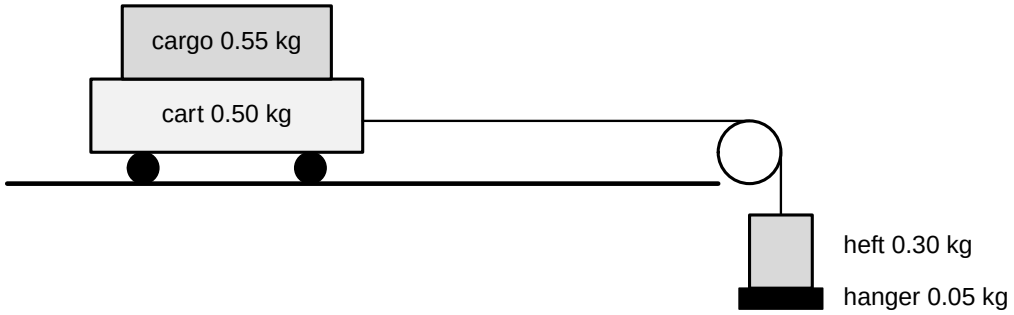
The total mass includes all four masses: $1.00 + 0.50 + 0.05 + 0.95 = 2.50$ kg.

The acceleration is the force divided by the total mass: $10 \div 2.5 = 4$ m/s^2.

- $(0.05 + 0.95) \times 10 = 10$ N
- $1.00 + 0.50 + 0.05 + 0.95 = 2.50$ kg
- $10 \div 2.5 = 4$ m/s^2

31. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.30 = 0.35$ kg, so the weight is $0.35 \times 10 = 3.5$ N.

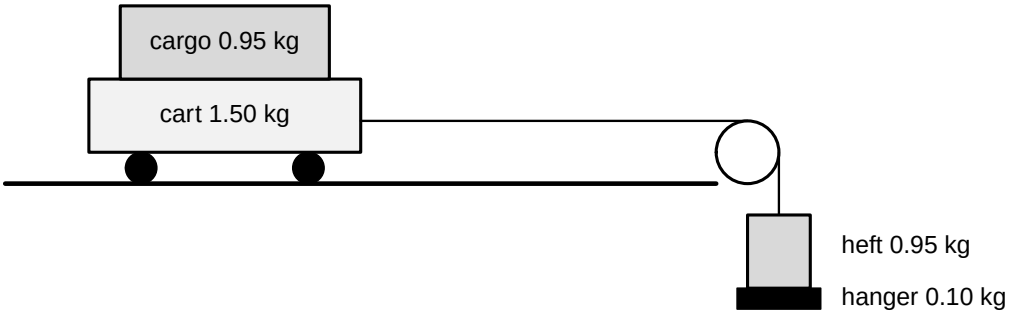
The total mass includes all four masses: $0.50 + 0.55 + 0.05 + 0.30 = 1.40$ kg.

The acceleration is the force divided by the total mass: $3.5 \div 1.4 = 2.5$ m/s^2.

- $(0.05 + 0.30) \times 10 = 3.5$ N
- $0.50 + 0.55 + 0.05 + 0.30 = 1.40$ kg
- $3.5 \div 1.4 = 2.5$ m/s^2

32. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.10 + 0.95 = 1.05$ kg, so the weight is $1.05 \times 10 = 10.5$ N.

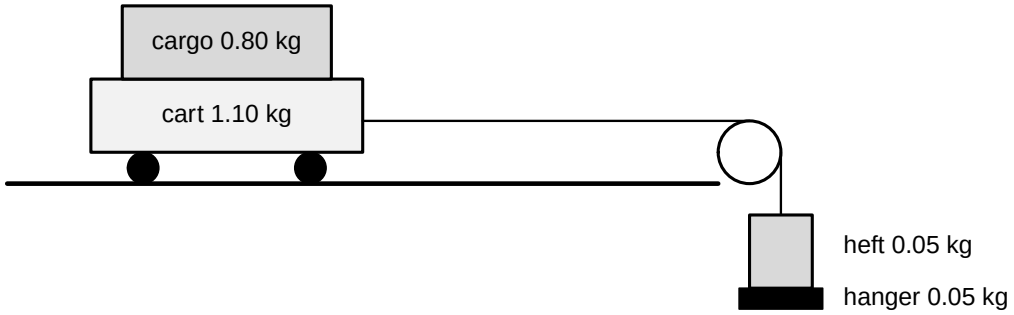
The total mass includes all four masses: $1.50 + 0.95 + 0.10 + 0.95 = 3.50$ kg.

The acceleration is the force divided by the total mass: $10.5 \div 3.5 = 3$ m/s^2.

- $(0.10 + 0.95) \times 10 = 10.5$ N
- $1.50 + 0.95 + 0.10 + 0.95 = 3.50$ kg
- $10.5 \div 3.5 = 3$ m/s^2

33. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.05 = 0.10$ kg, so the weight is $0.10 \times 10 = 1$ N.

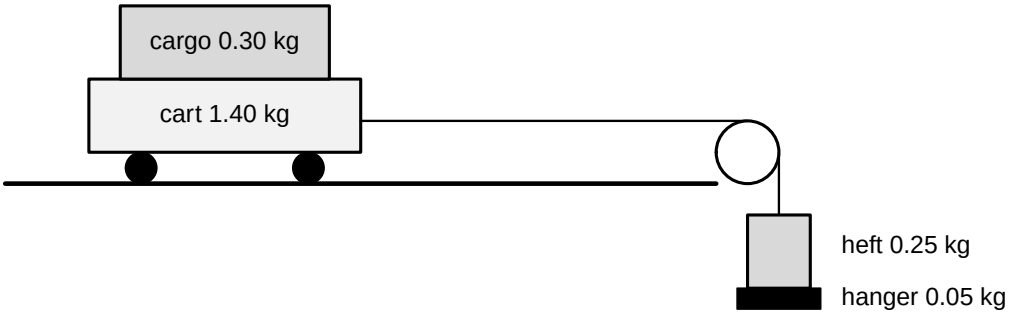
The total mass includes all four masses: $1.10 + 0.80 + 0.05 + 0.05 = 2.00$ kg.

The acceleration is the force divided by the total mass: $1 \div 2 = 0.5$ m/s^2.

- $(0.05 + 0.05) \times 10 = 1$ N
- $1.10 + 0.80 + 0.05 + 0.05 = 2.00$ kg
- $1 \div 2 = 0.5$ m/s^2

34. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.25 = 0.30$ kg, so the weight is $0.30 \times 10 = 3$ N.

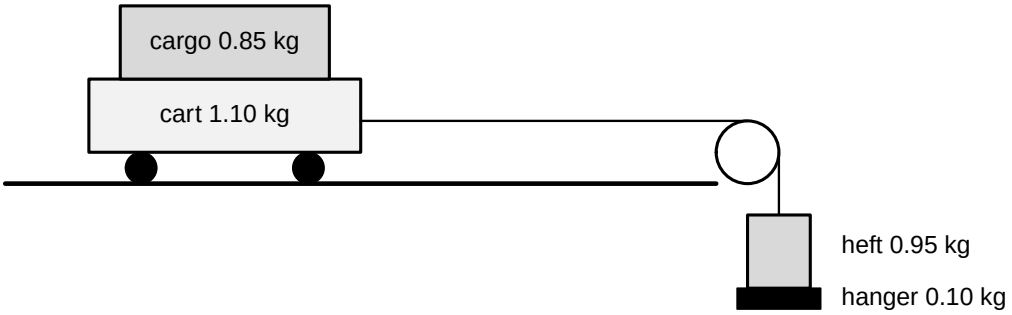
The total mass includes all four masses: $1.40 + 0.30 + 0.05 + 0.25 = 2.00$ kg.

The acceleration is the force divided by the total mass: $3 \div 2 = 1.5$ m/s^2.

- $(0.05 + 0.25) \times 10 = 3$ N
- $1.40 + 0.30 + 0.05 + 0.25 = 2.00$ kg
- $3 \div 2 = 1.5$ m/s^2

35. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.10 + 0.95 = 1.05$ kg, so the weight is $1.05 \times 10 = 10.5$ N.

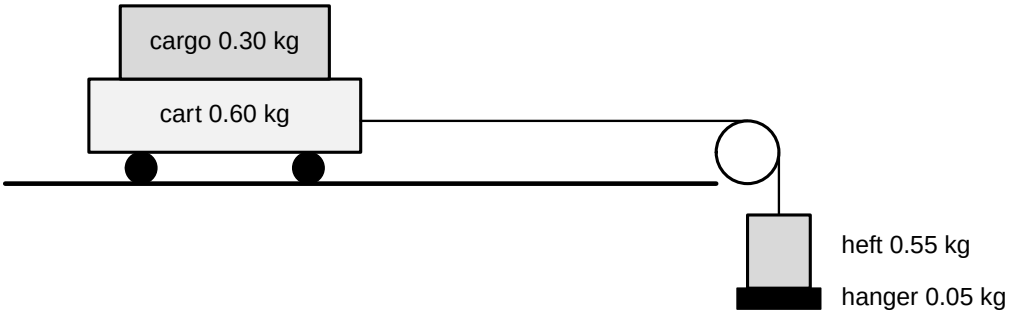
The total mass includes all four masses: $1.10 + 0.85 + 0.10 + 0.95 = 3.00$ kg.

The acceleration is the force divided by the total mass: $10.5 \div 3 = 3.5$ m/s^2.

- $(0.10 + 0.95) \times 10 = 10.5$ N
- $1.10 + 0.85 + 0.10 + 0.95 = 3.00$ kg
- $10.5 \div 3 = 3.5$ m/s^2

36. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.55 = 0.60$ kg, so the weight is $0.60 \times 10 = 6$ N.

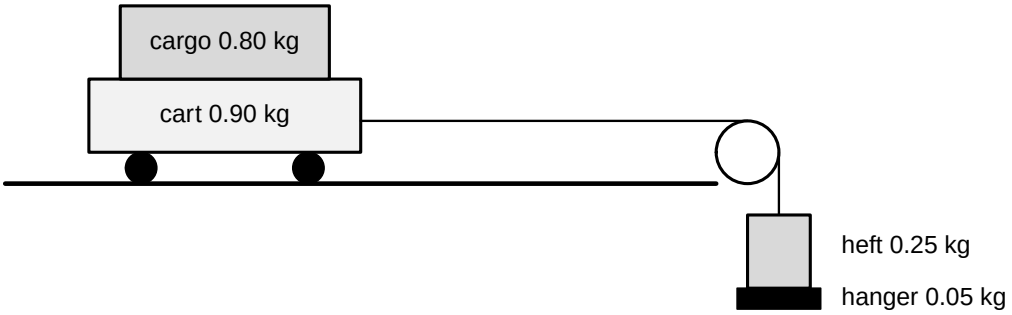
The total mass includes all four masses: $0.60 + 0.30 + 0.05 + 0.55 = 1.50$ kg.

The acceleration is the force divided by the total mass: $6 \div 1.5 = 4$ m/s^2.

- $(0.05 + 0.55) \times 10 = 6$ N
- $0.60 + 0.30 + 0.05 + 0.55 = 1.50$ kg
- $6 \div 1.5 = 4$ m/s^2

37. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.25 = 0.30$ kg, so the weight is $0.30 \times 10 = 3$ N.

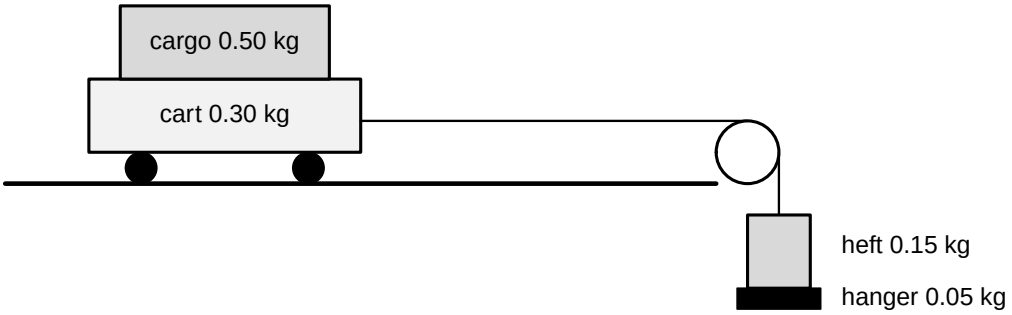
The total mass includes all four masses: $0.90 + 0.80 + 0.05 + 0.25 = 2.00$ kg.

The acceleration is the force divided by the total mass: $3 \div 2 = 1.5$ m/s^2.

- $(0.05 + 0.25) \times 10 = 3$ N
- $0.90 + 0.80 + 0.05 + 0.25 = 2.00$ kg
- $3 \div 2 = 1.5$ m/s^2

38. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.05 + 0.15 = 0.20$ kg, so the weight is $0.20 \times 10 = 2$ N.

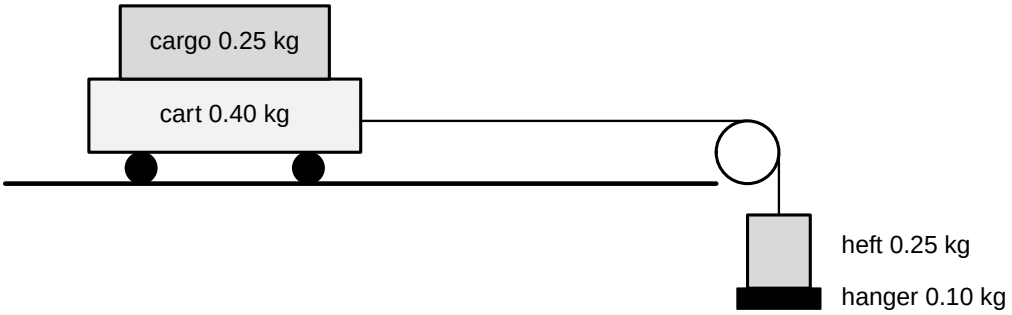
The total mass includes all four masses: $0.30 + 0.50 + 0.05 + 0.15 = 1.00$ kg.

The acceleration is the force divided by the total mass: $2 \div 1 = 2$ m/s^2.

- $(0.05 + 0.15) \times 10 = 2$ N
- $0.30 + 0.50 + 0.05 + 0.15 = 1.00$ kg
- $2 \div 1 = 2$ m/s^2

39. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.10 + 0.25 = 0.35$ kg, so the weight is $0.35 \times 10 = 3.5$ N.

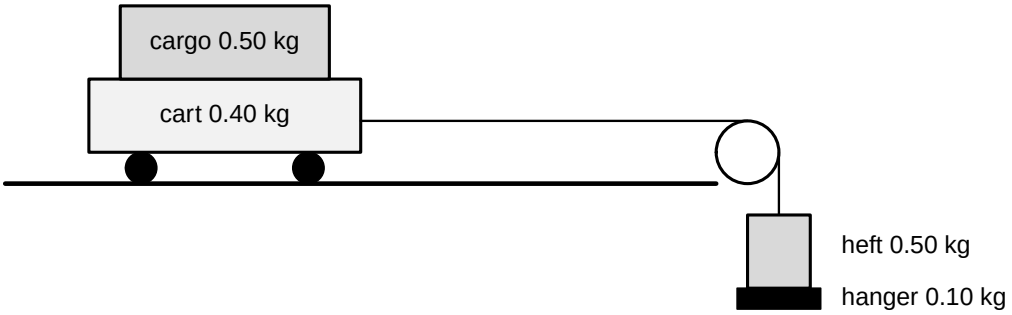
The total mass includes all four masses: $0.40 + 0.25 + 0.10 + 0.25 = 1.00$ kg.

The acceleration is the force divided by the total mass: $3.5 \div 1 = 3.5$ m/s^2.

- $(0.10 + 0.25) \times 10 = 3.5$ N
- $0.40 + 0.25 + 0.10 + 0.25 = 1.00$ kg
- $3.5 \div 1 = 3.5$ m/s^2

40. **Problem**

A cart is pulled along a level track by a string that runs over a pulley to a hanger. Some mass (cargo) rides on the cart, and some mass (heft) sits on the hanger. Ignore friction and the mass of the string.



- The **forward force** is the weight of the hanger plus the heft.
- Weight is found by multiplying mass by 10 N/kg.
- The **total mass** being pushed forward is made of all four masses: cart, cargo, hanger, and heft.

What is the forward force (include units!)?

What is the total mass (include units!)?

What is the acceleration of the cart (include units!)?

Solution

The forward force is the weight of the hanger plus the heft. Their total mass is $0.10 + 0.50 = 0.60$ kg, so the weight is $0.60 \times 10 = 6$ N.

The total mass includes all four masses: $0.40 + 0.50 + 0.10 + 0.50 = 1.50$ kg.

The acceleration is the force divided by the total mass: $6 \div 1.5 = 4$ m/s^2.

- $(0.10 + 0.50) \times 10 = 6$ N
- $0.40 + 0.50 + 0.10 + 0.50 = 1.50$ kg
- $6 \div 1.5 = 4$ m/s^2